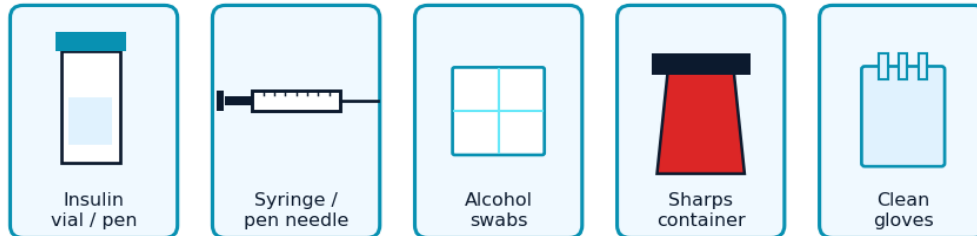


Insulin Injection Procedure

Standard operating procedure for safe subcutaneous insulin administration

Required Supplies



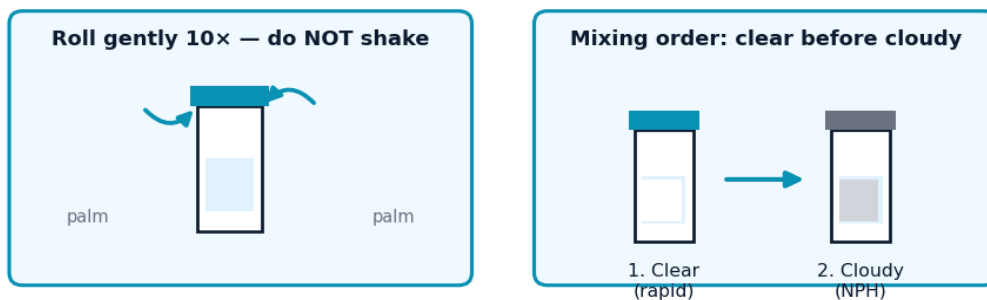
Step 1: Gather Supplies

Collect insulin vial or pen, appropriate syringe or pen needle (4–8mm), alcohol swabs, sharps container, and clean gloves. Verify the insulin type, concentration (U-100 or U-500), and expiration date.

■ **Never use expired insulin or cloudy rapid-acting insulin (except NPH which is normally cloudy).**

Note: Wash hands thoroughly with soap and water for at least 20 seconds before beginning.

Preparing the Insulin

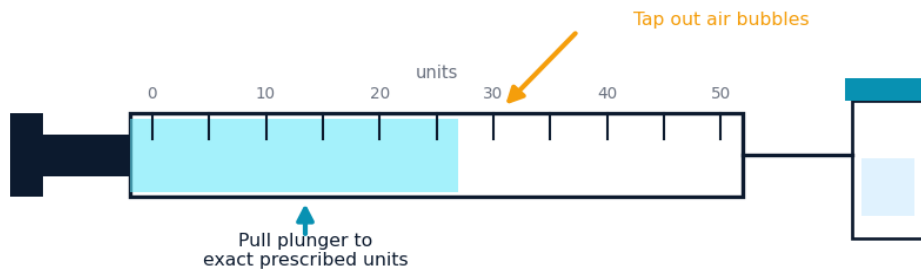


Step 2: Prepare the Insulin

For cloudy insulins (NPH, premixed), gently roll the vial between palms 10 times — do NOT shake. For clear insulins, inspect for particles. If using a vial, wipe the rubber stopper with an alcohol swab and let it dry for 15–30 seconds.

Note: If mixing insulins, always draw clear (rapid-acting) before cloudy (intermediate-acting) — 'clear before cloudy'.

Drawing Up the Correct Dose

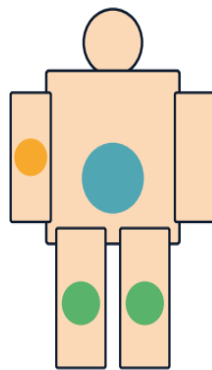


Step 3: Draw Up the Dose

Pull back the plunger to the prescribed number of units to draw air into the syringe. Insert the needle into the vial and push the air in. Invert the vial and draw back slightly more than the required dose. Remove air bubbles by tapping the syringe and pushing excess back into vial. Confirm the exact dose.

■ Double-check dose with a second nurse if administering to a hospitalized patient (per facility policy).

Subcutaneous Injection Sites & Absorption



Rotate sites systematically

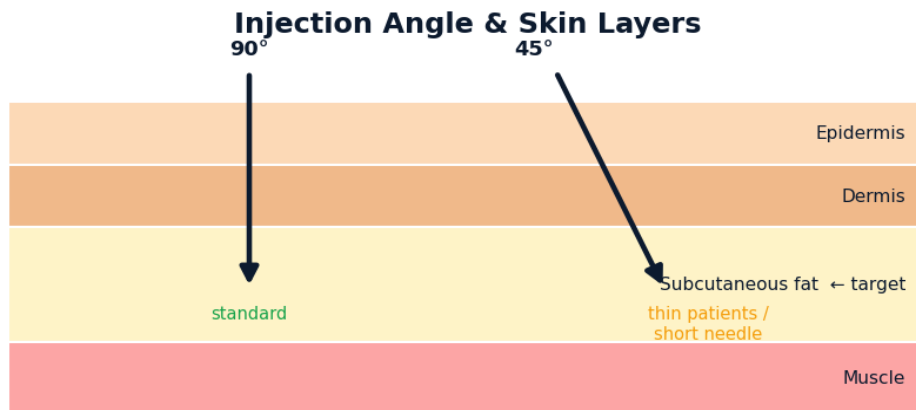
- Abdomen — fastest, most consistent (preferred for mealtime)
- Upper outer arm — moderate absorption
- Outer thigh / buttocks — slower absorption

Keep injections ≥ 2 inches (5 cm) from the navel and rotate to prevent lipohypertrophy.

Step 4: Select and Prepare Injection Site

Rotate injection sites systematically. Preferred sites: abdomen (fastest absorption, 2 inches from navel), outer thigh, upper outer arm, upper buttocks. Clean the site with an alcohol swab in a circular motion and allow to air-dry completely (10–30 seconds) — injecting into wet skin stings and may affect absorption.

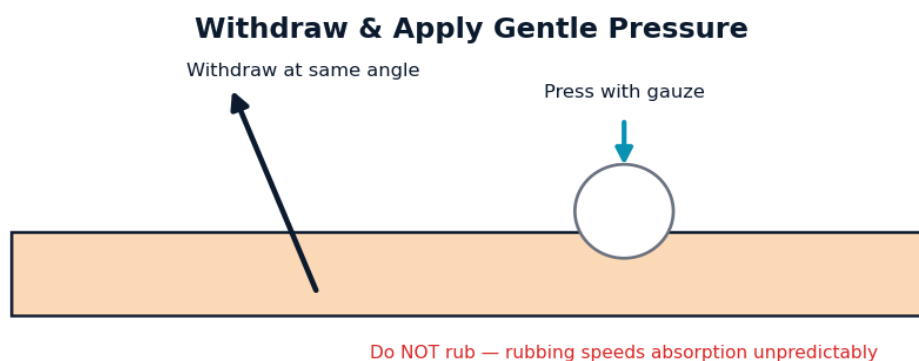
Note: Abdomen has the most consistent absorption rate and is preferred for meal-time insulin.



Step 5: Administer the Injection

Pinch a fold of skin (if using a longer needle or patient has little subcutaneous fat). Insert needle at 90° angle (45° for thin patients with short needles). Release skin fold. Inject insulin slowly and steadily. Hold for 10 seconds before withdrawing to prevent leakage.

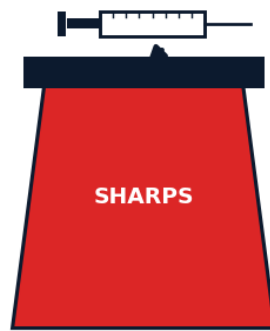
■ Do not inject into lumpy, scarred, or inflamed tissue — absorption will be erratic.



Step 6: Withdraw and Apply Pressure

Withdraw the needle at the same angle it was inserted. Apply gentle pressure with a clean cotton ball or gauze — do NOT rub (rubbing accelerates absorption unpredictably). If there is minor bleeding, hold pressure for 1–2 minutes.

Dispose Safely & Document



Never recap needles

Document

- ✓ Dose administered
- ✓ Time & site used
- ✓ Blood glucose reading
- ✓ Next rotation site

Step 7: Dispose and Document

Immediately dispose of the needle and syringe in an approved sharps container — never recap needles. Document the dose administered, time, injection site used, and patient's blood glucose reading. Schedule the next site for rotation.

■ Needles must NEVER be recapped, bent, or placed in regular trash — risk of needlestick injury.

Note: Monitor patient for hypoglycemia (rapid-acting insulin peaks in 1–3 hours).